

Introduction

This project analyzes historical baseball player data from the MLB to examine patterns in offensive performance through players' slugging percentage, how offense has changed over time, and what stats correlate with winning games. Analyses 1 and 2 use batting records and player demographics to determine which countries tend to produce the most powerful players and to examine the relationship between batting average and slugging. Analyses 3 and 4 explore changes in offense over time and speculate whether this reflects players improving or a shift in which team stats correlate with winning. These results will support decision-making for professional baseball teams in scouting and game strategy, and provide fans with interesting patterns in player performance.

[Link to Google Slides](#)

[Link to Project GitHub](#)

Dataset

The project dataset provides historical MLB data including player performance, revenue, awards, and team statistics. These categories of statistics made this dataset very suitable for many different kinds of analysis, allowing us to calculate advanced statistics and find important patterns in the data. The attributes that we took advantage of for this project included individual player statistics, demographic information, and team performances per game.

Analysis Technique

Analysis 1 used side-by-side boxplots to demonstrate whether the average slugging percentage by the player's country of birth had any significance. This technique provides the reader with an easily understandable graphic for the median and IQR of slugging percentage for each country. For 2, the focus is on exploring the value of slugging percentage against batting

average, which gives insight into a player's power through the use of a scatterplot providing a strong positive correlation between the two. For Analyses 3 and 4 the underlying idea is built on converting season totals into per-game averages to account for different eras and schedule lengths. To show how offense has changed over time, per-game rates were plotted using a line chart to easily view the changes year to year. This made it easy to spot drastic dips or jumps, like the jump in home runs per game around 1920, or the steady rise in strikeouts. For the question of what teams' stats predict winning, a (Pearson) correlation coefficient was calculated between each team statistic used and the team's winning percentage. Correlation is an appropriate technique since both the winning percentage and the potential indicator are continuous variables that are measured at the same time and level, being one value per team per season. This provides a fairly interpretable measure by condensing multiple scatterplots into a matrix to compare the values against each other.

Results

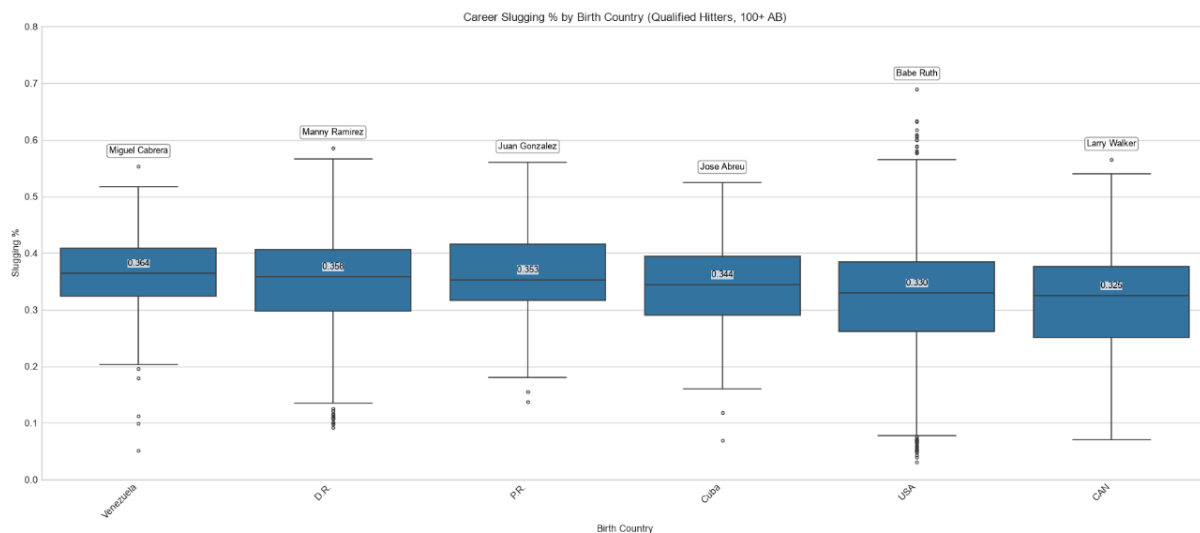


Figure 1: The side-by-side boxplots suggest that certain countries may produce more powerful hitters, as there is a fairly significant difference in slugging between the highest (Venezuela) and lowest (Canada) countries' median and 3rd quartile. This information could be

utilized by team scouts who are looking for a powerful type of player to recruit to their team, however much more data would need to be collected for these results to have significant meaning.

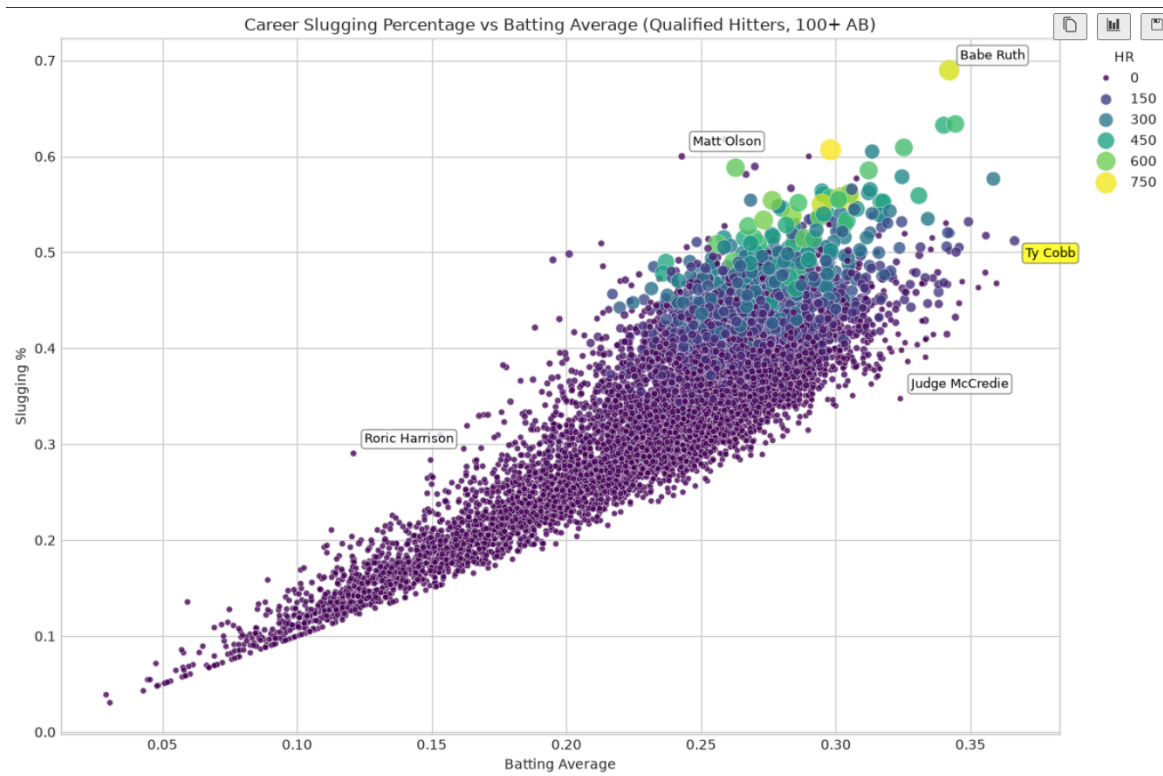


Figure 2: The scatterplot of batting average vs slugging percentage shows a strong, positive relationship between the two stats. More importantly, this scatterplot highlights players who stand out as being exceptional in a certain playstyle, like high power and low contact or high contact and low power. Players like Judge McCredie stand out with an extremely high batting average but mediocre slugging, while Matt Olson has an exceptional slugging percentage but a below-average batting average. Having this information on their players is vital for the strategy of a team, specifically in setting their batting order. Teams put high-contact hitters in the 1-3 spots in the batting order, followed by power hitters to drive home the contact hitters.

League-Wide Offensive Rates by Season, 1901-2017

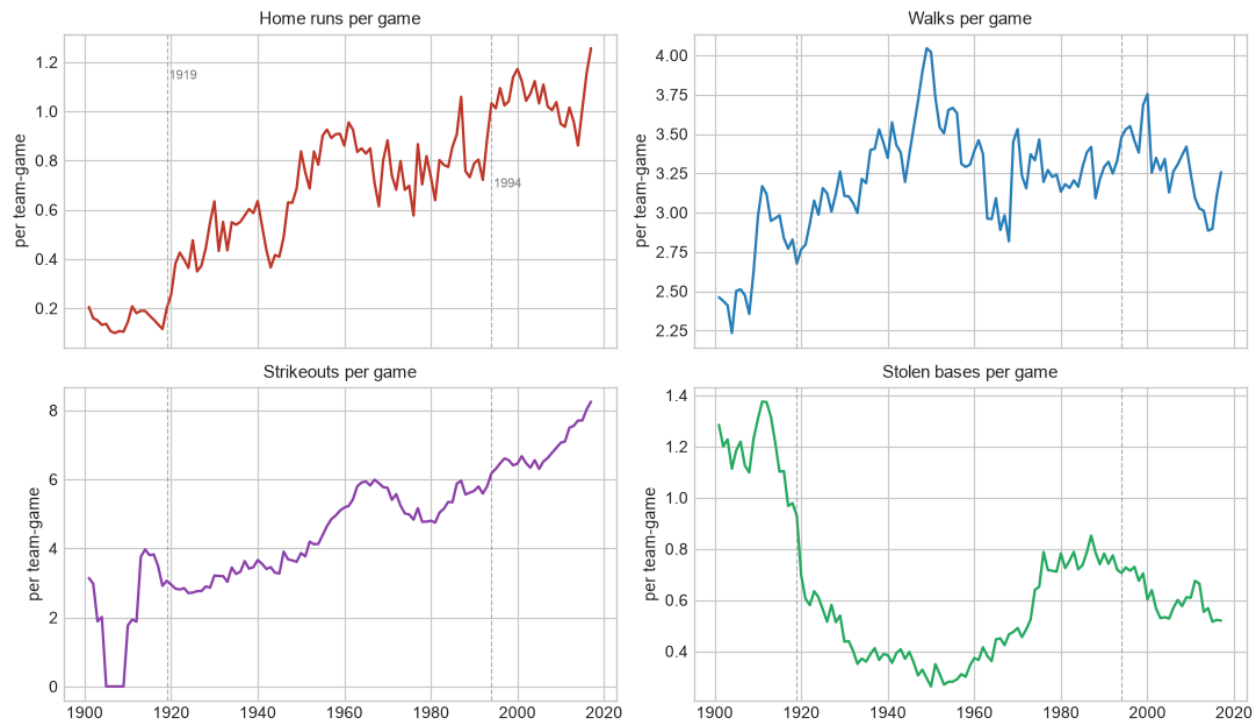


Figure 3: League wide home runs, walks, strikeouts, and stolen bases per team game.

The series of offensive stats over time shows that home runs per game increased more than six times the initial value from 1901. With home runs per game as a focus, the most notable jumps being around 1920 or the start of the “live ball” era and the mid-90s commonly referred to as the steroid era. The steady climb of strikeouts climbed similarly with home runs per game, and the dip of stolen bases can be viewed as almost a “go big or do down trying” type of strategy, which has been the common process over time.

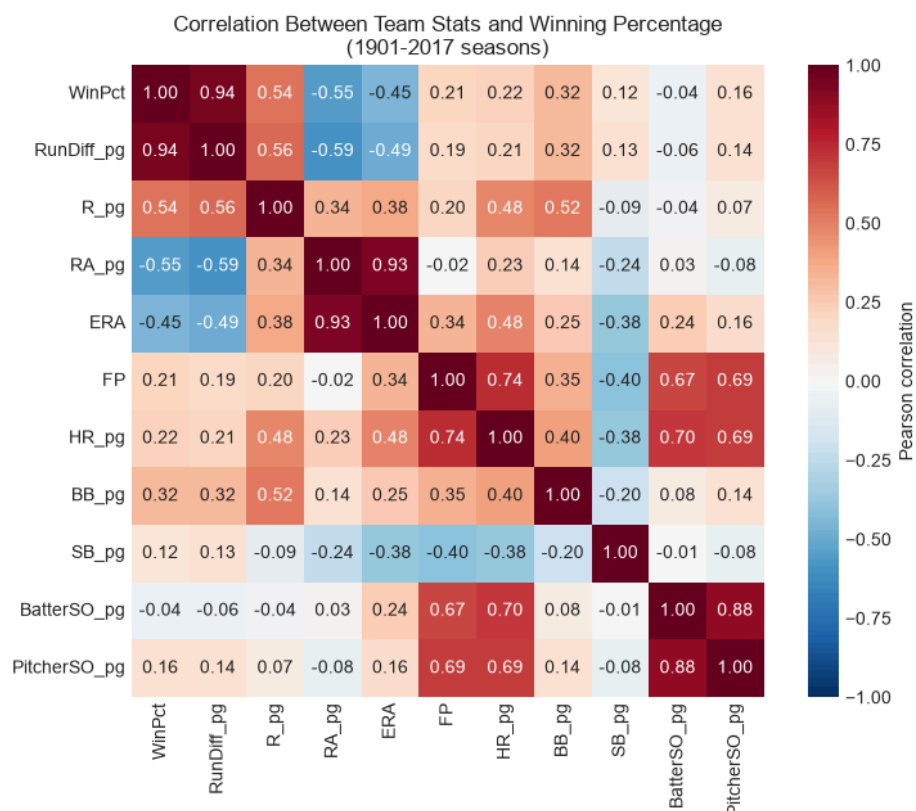


Figure 4: Correlation matrix between team season statistics and winning percentage. The analysis of the correlations shown in Figure 4, found that a team's run differential per game correlates with winning percentage at .94 across 2,482 teams. This can be taken as, the rate at which a team wins by a large margin the more likely they are to win a game. There are a few statistics that are tied to "small ball" strategies that appear to be weaker than expected. This is also reflected in the changes in offense over time in Figure 3, where teams lean toward a tolerance for high strikeout, high power hitters being more defensible than the small ball strategy traditionally preached.

Technical

In order to calculate batting average and slugging percentage from the batting dataset, we grouped batting data by playerID, then divided total bases by at bats to calculate slugging, and divided hits by at bats to calculate batting average. The data was filtered to only include players

with at least 100 at bats, removing players with missing or invalid birth countries, and limiting the comparison to only countries with 100 qualified hitters. Analyses 3 and 4 filtered out seasons before 1901 because of rule changes, schedule lengths and some inconsistencies with foul balls not always counting as strikes. Side-by-side box plots and scatter plots with labels suitably displayed the results found from the investigation in a clear way for any general audience. Box plots allow for clear comparison between categories, while a labeled scatterplot clearly demonstrates trends and interesting outliers. If these analyses were to be extended or done with a different approach, they might include additional control/qualifying stats like league era or ballpark dimensions to provide further context. Every counting statistic was converted into a per-game rate for the correlation to winning. For the change in offense, year to year this was done by summing the number of games separately across every team in a season and then dividing the two totals. This was done instead of averaging each team's individual per-game rate since there are a small number of teams that played shortened seasons.